Northstar Retail Analytics

Customer Retention and Churn Risk Review

DEMO REPORT - ALL NAMES, ORGANIZATIONS, DATA, AND RESULTS ARE FICTIONAL

Report version: 1.2

Prepared: 18 September 2025

Reporting period: 1 January 2025 to 30 June 2025

Purpose

This sample report describes a fictional retail subscription business and a customer churn analytics project. It is designed as a long, searchable example document for testing retrieval-augmented generation (RAG) systems. It contains sections, definitions, tables, dates, metrics, decisions, risks, and operational procedures so that a question-answering system can be evaluated on different question types.

Suggested test questions appear at the end of the report. Answers should be grounded in the text, and the system should say when a requested fact is not specified.

1. Executive Summary

Northstar Retail Analytics is a fictional online retailer that sells household essentials, personal care items, and reusable home products through a paid delivery membership. Its leadership team commissioned a customer retention review after the monthly cancellation rate increased during the first half of 2025.

The analytics team developed a churn risk workflow to help customer support and retention teams identify accounts that may cancel within the next 30 days. The workflow combines activity data, membership information, service history, and a supervised classification model. It produces a risk score and a small set of explanation fields for analyst review; it does not automatically cancel, discount, suspend, or otherwise change a customer account.

Headline findings

- The validated evaluation dataset contained 24,000 customer accounts, including 3,120 accounts that cancelled within the defined prediction window.
- The selected gradient-boosted tree model achieved 0.81 area under the receiver operating characteristic curve (ROC AUC) and 0.68 precision at the approved decision threshold.
- The approved threshold was a churn probability of 0.62. Accounts above that threshold are placed in a review queue; an analyst makes the final decision about any outreach.
- The strongest predictive signals were declining order frequency, unresolved support cases, failed payment attempts, and membership tenure under 90 days.
- The initial pilot was planned for 6 October 2025, with a four-week monitoring period and a formal review on 7 November 2025.

Key figures at a glance

Measure	Value	Definition
Reporting period	1 Jan - 30 Jun 2025	Period used for operational trend reporting
Evaluation accounts	24,000	Unique accounts in the held-out evaluation set
Cancellations in evaluation set	3,120	Accounts cancelled within the 30-day target window
Selected model ROC AUC	0.81	Ranking ability across positive and negative outcomes
Approved review threshold	0.62	Predicted churn probability required for review
Pilot start	6 Oct 2025	First planned operational use

Interpretation note: results in this report are fictional and are provided only to test a document QA system. They are not real business performance claims.

2. Organization and Business Context

Northstar Retail Analytics operates the fictional Northstar Home Club membership program. Members pay a monthly fee for scheduled delivery, reorder reminders, and access to a curated catalog of household products. The company serves customers in three fictional regions: Lakeside, Hillview, and Westport.

The business has four main operating groups. The Commerce group manages the catalog and checkout. The Member Experience group manages onboarding, support, and retention outreach. The Data Products group maintains reporting pipelines and predictive services. The Risk and Privacy group reviews data use, access permissions, and customer communications.

Business problem

During Q1 2025, voluntary cancellations averaged 8.4% of active members per month. In Q2 2025, the average rose to 10.1%. Exit survey comments most often mentioned delivery reliability, unclear value from the membership fee, and unresolved support issues. The leadership team requested an analytics workflow that could help analysts prioritize outreach without treating a prediction as a certainty.

Business goals

- Prioritize potentially at-risk accounts for human review before the member reaches the cancellation page.
- Understand which observable behaviors are associated with later cancellation.
- Measure model performance and outreach outcomes separately from overall cancellation trends.
- Avoid unfair or unnecessary intervention by reviewing model errors across customer groups and regions.
- Give support analysts a concise explanation of why an account entered the review queue.

Out of scope

The workflow does not set prices, determine eligibility for credit, automatically send marketing messages, make employment decisions, or infer sensitive personal characteristics. It is not intended to replace customer support judgment or the existing cancellation process.

Vocabulary used in this report

Term	Meaning
Churn	A membership cancellation recorded during the defined outcome window.
At-risk account	An account whose predicted churn probability is at or above the review threshold.
Review queue	A list of accounts that an authorized analyst can inspect before deciding whether outreach is appropriate.
Prediction window	The 30 days after the scoring date.
Outcome window	The period used to determine whether cancellation occurred after scoring.

3. Project Scope, Deliverables, and Timeline

The project was delivered as a prototype-to-pilot workflow. The first objective was to confirm that the data could be used to form a reliable prediction target. The second was to compare several candidate models. The third was to package the selected model behind a validated scoring API and document a safe human-review process.

Project deliverables

- A versioned data preparation pipeline with checks for missing values, duplicate records, and feature leakage.
- A baseline model comparison report with class-sensitive metrics and an agreed review threshold.
- A model artifact and preprocessing configuration stored with a version identifier.
- A prediction API that validates required fields and returns a risk score with a model version.
- An analyst review queue prototype and an operating guide describing review, escalation, and feedback capture.
- A monitoring dashboard specification for score distribution, data quality, API health, and observed outcomes.

Timeline

Date	Milestone	Status in this fictional report
6 Jan 2025	Project kickoff and stakeholder interviews	Completed
31 Jan 2025	Data inventory and target definition approved	Completed
21 Feb 2025	Data quality checks and initial feature list	Completed
28 Mar 2025	Baseline model comparison	Completed
25 Apr 2025	Threshold review and error analysis	Completed
23 May 2025	API prototype and request validation	Completed
20 Jun 2025	Pilot operating guide and risk review	Completed
6 Oct 2025	Planned start of controlled pilot	Planned
7 Nov 2025	Formal pilot review and go/no-go decision	Planned

Acceptance criteria

The pilot may begin only after all of the following are met: the evaluation metrics are signed off by the Analytics Lead; the Risk and Privacy reviewer confirms the permitted fields; API authentication is enabled; the rollback procedure has been tested; and the Member Experience manager has confirmed analyst staffing for the first four weeks.

4. Data Sources and Data Dictionary

The development dataset was assembled from four fictional internal systems. The extraction snapshot was taken on 3 July 2025. Only records with a valid account identifier and a usable scoring date were eligible for modeling. Personal message content was not included; support interactions were represented through category and resolution fields only.

Source	Fields used	Refresh cadence	Primary owner
Order Ledger	account_id, order_date, order_value, order_status	Daily	Commerce Operations
Membership Service	join_date, plan_type, renewal_date, account_status	Daily	Member Experience
Support Tracker	case_open_date, case_category, resolution_status, resolution_hours	Hourly	Support Operations
Payment Events	payment_attempt_date, payment_result, retry_count	Daily	Billing Operations

Important fields

Field	Type	Description	Validation rule
account_id	String	Pseudonymous unique account key	Required; unique per account table
orders_30d	Integer	Completed orders during the 30 days before scoring	Must be 0 or greater
order_change_60d	Float	Relative change in order count against the preceding 60-day period	Bounded to -1.0 through 5.0 after clipping
open_cases	Integer	Support cases unresolved at scoring time	Must be 0 or greater
median_resolution_hours	Float	Median resolution duration over the previous 90 days	Missing if no resolved case exists
failed_payments_30d	Integer	Failed payment attempts in the prior 30 days	Must be 0 or greater
tenure_days	Integer	Days between join date and scoring date	Must be 0 or greater
plan_type	Category	Basic, Plus, or Family membership tier	Unknown values mapped to Other
churn_30d	Binary target	1 if cancellation occurred in the next 30 days; otherwise 0	Used only as the supervised learning target

Data snapshot summary

The full raw extract contained 31,480 account-period rows. After duplicate removal, eligibility checks, and exclusion of records without a complete target window, 28,760 rows remained for development and evaluation. The held-out evaluation set described in Section 7 contains 24,000 account records drawn from a later time period. These counts describe separate processing stages and should not be added together.

5. Data Preparation and Leakage Controls

The data preparation pipeline was implemented as a versioned sequence of transformations. Each transformation was fitted only on training data where fitting was necessary, then applied to validation and evaluation data using the same configuration. The pipeline retained an audit log of row counts and records excluded at each stage.

Preparation steps

- Removed duplicate account-period rows using account_id and scoring_date as the composite key.
- Standardized timestamps to UTC before calculating rolling windows.
- Replaced invalid negative order counts and retry counts with missing values, then sent affected records to a data-quality report.
- Imputed numerical missing values using medians learned from the training partition.
- Mapped unseen categories to an Other category rather than failing inference.
- One-hot encoded categorical fields and scaled numerical fields for models that required scaling.
- Clipped extreme order-change values at the documented bounds to reduce sensitivity to isolated data errors.
- Stored the feature schema alongside the model artifact so the API could reject unexpected fields.

Leakage controls

A feature was considered leaked if it used information that would not be available at the scoring time or if it directly reflected a cancellation that occurred after scoring. The team excluded cancellation reason, final account status after the outcome window, refunds issued after cancellation, and any support notes written after the scoring timestamp. The churn_30d target was never included among the input features.

Time-based split design

Partition	Scoring dates	Purpose
Training	1 Jan - 31 Mar 2025	Fit preprocessing and candidate models
Validation	1 Apr - 30 Apr 2025	Compare models and tune thresholds
Evaluation	1 May - 30 Jun 2025	Final metrics on later unseen observations

Why a time-based split was chosen

Randomly distributing all months across partitions could allow future behavior patterns to influence the training data. A time-based design better reflects the intended use case: score current accounts based on prior activity and evaluate whether those scores predict a subsequent cancellation window.

6. Modeling Approach

The team compared three candidate models: Logistic Regression, Random Forest, and Gradient-Boosted Trees. Logistic Regression was used as an interpretable baseline. Random Forest provided a bagged tree comparison. Gradient-Boosted Trees were included to model non-linear relationships and interactions between activity signals.

Model	Strengths	Trade-offs	Role in comparison
Logistic Regression	Fast; coefficients are relatively easy to inspect	May underfit complex interactions	Interpretable baseline
Random Forest	Handles non-linear patterns; robust to many feature interactions	Can be large; probability calibration may need attention	Tree-based benchmark
Gradient-Boosted Trees	Strong ranking performance on tabular features	Needs careful tuning and monitoring	Selected candidate

Training configuration

- The target was cancellation within 30 days of the scoring date.
- Class weights were tested because cancellation was less common than retention.
- Hyperparameters were selected using validation data only.
- The decision threshold was selected after reviewing precision, recall, queue capacity, and the cost of unnecessary outreach.
- Calibration and subgroup stability were reviewed before recommending the pilot.

Evaluation measures

Metric	Question it answers	Limitation
ROC AUC	Can the model rank a cancelling account above a retained account?	Does not determine the operational threshold by itself.
Precision	Among flagged accounts, how many cancelled within the target window?	Can fall as more accounts are flagged.
Recall	Among all accounts that cancelled, how many were flagged?	Higher recall may create a larger review queue.
F1 score	What is the harmonic balance of precision and recall?	May not reflect the business cost of false positives and false negatives.
Brier score	How close are predicted probabilities to observed outcomes?	Needs interpretation alongside calibration plots and prevalence.

Model outputs are prioritization signals, not facts about an individual customer. Analysts must review the supporting activity and account context before deciding whether to contact a member.

7. Evaluation Results and Threshold Decision

The held-out evaluation set contains 24,000 account records. Of these, 3,120 accounts cancelled within the target window, giving an observed cancellation prevalence of 13.0%. The comparison below reports the same test population for each candidate model.

Model	ROC AUC	Precision at threshold	Recall at threshold	F1 score
Logistic Regression	0.74	0.54	0.61	0.57
Random Forest	0.78	0.62	0.55	0.58
Gradient-Boosted Trees	0.81	0.68	0.59	0.63

The Gradient-Boosted Trees model was selected because it achieved the highest ROC AUC and the strongest precision among the compared candidates at the reviewed operating points. The selection was not based on ROC AUC alone; the team also checked the expected queue volume and the types of errors produced.

Approved threshold

The approved threshold is 0.62 predicted churn probability. An account is flagged for review when its score is greater than or equal to 0.62. Accounts below the threshold are not automatically marked as safe; they simply do not enter this particular priority queue.

Threshold	Accounts flagged	Precision	Recall	Operational comment
0.50	4,480	0.55	0.79	High queue volume; more false positives
0.62	2,710	0.68	0.59	Approved balance for pilot staffing
0.75	1,490	0.77	0.37	Smaller queue; misses more eventual cancellations

Error analysis

False positives include accounts that were flagged but did not cancel during the next 30 days. Some may still have benefited from support, while others would have received unnecessary outreach. False negatives include accounts that cancelled but were not flagged. The team recommends reviewing a sample of both error types every week during the pilot and documenting whether each error resulted from data quality, behavior change, timing, or a limitation of the model.

The reported values are fictional evaluation figures for demo purposes. They should not be interpreted as a forecast of real-world performance.

8. Behavior Signals and Segment Findings

The model review identified several recurring signals associated with higher cancellation risk. A signal can be predictive without being causal. The team did not interpret these relationships as proof that a particular behavior caused a customer to cancel.

Signal	Observed pattern	How an analyst should interpret it
Declining order frequency	Accounts with a sharp reduction in completed orders were more likely to cancel.	Check whether the member changed routine, moved, or experienced a fulfillment issue.
Unresolved support cases	Risk increased when cases remained open beyond the normal resolution time.	Review the open case and ensure a support resolution is underway.
Failed payment attempts	Repeated failures were associated with increased risk.	Check payment status and use approved billing support procedures.
Short membership tenure	Accounts with less than 90 days of tenure were more likely to cancel.	Review onboarding, first-order success, and membership-value clarity.
Repeated delivery delay	Two or more delayed deliveries in 30 days were associated with elevated risk.	Verify the delivery record before outreach.

Regional monitoring

Region	Evaluation accounts	Observed cancellation rate	Review note
Lakeside	9,600	12.1%	Largest region; stable data completeness
Hillview	7,200	13.8%	Higher incidence of delivery-delay cases
Westport	7,200	13.3%	Payment failure fields had more missing values before cleanup
Total	24,000	13.0%	Weighted overall rate

The region-level rates are descriptive, not evidence of inherent customer differences. In Westport, the team must monitor missing payment data because missingness can affect score quality. In Hillview, operational delivery metrics should be reviewed alongside model output to avoid treating a service failure as a customer problem.

Recommended analyst review checklist

- Confirm that the score was produced from the latest valid data snapshot.
- Read the displayed feature explanations and verify them against the account record.
- Check for an open unresolved support issue before any retention message is drafted.
- Use only approved outreach templates and record the decision in the review system.
- Escalate incorrect account data or suspected model errors to the Analytics Support queue.

9. Technical Architecture and API Contract

The prototype has five logical components: an ingestion job, a feature pipeline, a model scoring service, a review queue, and a monitoring layer. Batch ingestion runs each morning at 05:30 UTC. The scoring service exposes a private HTTPS endpoint to authorized internal clients. The endpoint does not return customer message text or other unneeded personal content.

Scoring flow

- The ingestion job validates the latest source extracts and publishes a versioned feature snapshot.
- The feature pipeline calculates the approved rolling-window features and applies the stored preprocessing configuration.
- The scoring API checks the request schema and verifies the model version is active.
- The model returns a churn probability between 0 and 1 plus a model version identifier.
- The review queue applies the 0.62 threshold and displays the highest-priority accounts to authorized analysts.
- The monitoring layer records system health and appropriate outcome aggregates without storing unnecessary request payloads.

Example request

The following is a fictional example of a request body. Values are illustrative and do not refer to a real person.

```
{
  "account_id": "DEMO-10482",
  "orders_30d": 1,
  "order_change_60d": -0.50,
  "open_cases": 1,
  "failed_payments_30d": 0,
  "tenure_days": 74,
  "plan_type": "Plus"
}
```

Example response

```
{
  "risk_probability": 0.67,
  "review_required": true,
  "model_version": "churn-gbt-1.2.0",
  "scored_at_utc": "2025-10-06T05:42:13Z"
}
```

API validation rules

Rule	Expected behavior
Missing required field	Return HTTP 400 with a non-sensitive validation message.
Unknown plan_type	Map to Other if allowed by schema; otherwise reject the request.
Negative count field	Reject request and log a validation event.
Unauthorized client	Return HTTP 401 or 403 and do not score.
Model unavailable	Return HTTP 503; the review queue should not treat failure as a low-risk score.

10. Operational Runbook and Service Levels

The pilot team is expected to use the workflow during business hours. It is not a real-time customer decision service. If the daily batch is delayed, the previous score set must be marked stale and must not silently be presented as a new result.

Service objectives

Service measure	Target	Action if target is missed
Daily batch completion	By 06:00 UTC on business days	Alert Analytics Support if not complete by 06:15 UTC
Scoring API availability	99.5% during pilot business hours	Record incident and use the documented fallback process
Single request latency	p95 below 800 milliseconds	Check load, feature service, and model service timings
Feature freshness	Latest complete snapshot less than 30 hours old	Mark scores stale and suspend new review queue publication
Critical incident acknowledgement	Within 30 minutes during support hours	Escalate to the Data Products Lead

Incident categories

- Severity 1: unauthorized data exposure, incorrect bulk action, or scoring results delivered to an unauthorized client. Immediately disable the affected endpoint and notify the Security Lead.
- Severity 2: incorrect feature generation, repeated score failures, or stale data affecting the review queue. Pause publication of new scores and notify the Analytics Lead.
- Severity 3: isolated validation errors or dashboard display defects that do not affect the underlying model output. Log a ticket and fix during normal support hours.

Rollback procedure

The Data Products Lead may roll back to the last approved model artifact if a critical metric degrades or a defect is confirmed. The operator must disable publication of new scores, restore the previous artifact and configuration, run a validation sample, confirm API health, and document the rollback time and approver. Historical analyst decisions must not be silently overwritten.

Ownership and escalation

Role	Accountability
Analytics Lead	Model evaluation, feature definitions, threshold approval
Data Products Lead	API reliability, deployments, model artifact versioning
Risk and Privacy Lead	Permitted data use, access review, incident escalation
Member Experience Manager	Analyst training, queue procedures, outreach templates
Analytics Support	Daily batch checks, triage, data-quality tickets

11. Privacy, Governance, and Risk Controls

The workflow is designed for decision support and must follow data minimization principles. The model should use only approved operational fields that are necessary for retention analysis. Source data access is role-based and logged. The model does not ingest full free-text support conversations, payment card numbers, passwords, government identifiers, or precise location history.

Risk register

Risk	Likelihood	Impact	Mitigation	Owner
Feature leakage	Medium	High	Time-based split, leakage checklist, feature review	Analytics Lead
Data drift	Medium	Medium	Weekly feature distribution and score distribution checks	Data Products
Unnecessary outreach	Medium	Medium	Human review, approved templates, feedback logging	Member Experience
Uneven data quality by region	Medium	High	Report missingness and evaluate subgroup metrics	Analytics Lead
Unauthorized access	Low	High	Private endpoint, role-based access, audit log	Risk and Privacy
Misinterpretation of score	Medium	High	Training, clear labels, no automatic action	Member Experience

Human oversight requirements

- A score cannot trigger an automatic price change, cancellation, or account suspension.
- An analyst must review the evidence before any retention outreach is initiated.
- A customer must be handled under existing support policies regardless of model score.
- Analysts can mark a score as not useful and select a reason from the feedback list.
- The Analytics Lead reviews false-positive and false-negative samples every week during the pilot.

Data retention

Raw staging extracts are retained for 30 days. Feature snapshots used for scoring are retained for 90 days. Aggregated model monitoring data is retained for 13 months. Access logs are retained for 180 days. A legal hold or a documented security investigation may require a different schedule approved by the Risk and Privacy Lead.

Model limitations

The model can miss sudden life changes, one-time service failures, or cancellations that are not reflected in historical behavior. The target definition covers recorded cancellations within 30 days and does not capture every reason a member may disengage. A high score does not mean that a customer will definitely cancel, and a low score does not mean that a customer has no unmet need.

12. Monitoring Plan and Future Roadmap

Monitoring is split into data quality, model behavior, operational reliability, and business outcomes. These categories are reviewed separately so that a change in cancellation rates is not automatically attributed to the model.

Monitor	Frequency	Alert threshold	First response
Missing required features	Each batch	More than 2% of eligible rows	Pause publication and investigate the source
Feature distribution drift	Weekly	Population Stability Index above 0.20 for a monitored feature	Inspect source changes and assess model impact
Score distribution shift	Weekly	Mean risk score changes by more than 20% from the four-week baseline	Check data, pipeline version, and seasonality
Precision on matured outcomes	Monthly	Below 0.60 for two consecutive reports	Review threshold and error samples
API error rate	Daily	Above 1% over a rolling hour	Inspect service health and dependencies
Regional missingness gap	Weekly	More than 5 percentage points between regions	Investigate source process and pause affected use if needed

Roadmap

Phase	Target date	Planned work	Exit condition
Phase 1: Controlled pilot	Oct-Nov 2025	Train analysts, monitor queue quality, review errors weekly	Four weeks of stable operations and documented findings
Phase 2: Evaluation	Dec 2025	Compare flagged-account outcomes and collect feedback	Approved evaluation memo
Phase 3: Workflow refinement	Jan 2026	Improve explanations, queue filters, and stale-data handling	User acceptance test passed
Phase 4: Broader rollout decision	Feb 2026	Review capacity, controls, fairness checks, and support readiness	Leadership go/no-go decision

The project will not automatically progress from pilot to wider rollout. The February 2026 decision requires evidence that the workflow is useful to analysts, the data remains fit for purpose, and the controls are operating as designed.

13. Frequently Asked Questions

What does the model predict?

It predicts the probability that an account will cancel within 30 days after the scoring date.

Which model was selected?

Gradient-Boosted Trees was selected, with an evaluation ROC AUC of 0.81.

What is the approved threshold?

The approved review threshold is 0.62. Scores at or above 0.62 enter the analyst review queue.

Does the system automatically contact customers?

No. An authorized analyst must review the account and decide whether outreach is appropriate.

What was the evaluation sample size?

The held-out evaluation set contained 24,000 accounts, including 3,120 cancellations in the target window.

When is the pilot planned to start?

The pilot is planned to start on 6 October 2025.

When is the go/no-go decision?

The formal pilot review is scheduled for 7 November 2025, followed by a broader rollout decision planned for February 2026.

Which source contains payment outcomes?

The Payment Events source contains payment attempt dates, payment results, and retry counts.

What happens when the score API is unavailable?

The service returns an error, and the review queue must not treat the failure as a low-risk score.

How long are raw staging extracts retained?

Raw staging extracts are retained for 30 days.

What is the critical risk related to model use?

A key risk is that an analyst may mistake a probability for certainty. Human review and a prohibition on automatic account actions are required controls.

What should happen if feature freshness exceeds 30 hours?

The scores must be marked stale and new review queue publication must be suspended.

14. Glossary and Suggested RAG Test Questions

Term	Definition
AUC	Area under the receiver operating characteristic curve; a ranking metric.
Calibration	Agreement between predicted probabilities and observed frequencies.
Data leakage	Use of information during training that would not be available at the intended scoring time.
Embedding	A vector representation of text or another item used for similarity search.
FAISS	A library for efficient similarity search over dense vectors.
False negative	An account that cancels within the target window but was not flagged at the threshold.
False positive	An account that was flagged but does not cancel within the target window.
Human review	A process in which an authorized person inspects evidence before taking an action.
RAG	Retrieval-Augmented Generation: retrieving relevant material and using it as context for generated answers.
ROC AUC	A measure of how well a model ranks positive examples above negative examples across thresholds.
TF-IDF	A text weighting method that emphasizes terms useful for distinguishing documents.

Suggested evaluation questions

- What is the name of the fictional membership program?
- What reporting period is covered by this report?
- How many accounts were in the held-out evaluation set?
- How many cancellations occurred in the evaluation set, and what was the prevalence?
- Which model was selected and what ROC AUC did it achieve?
- What is the approved threshold and what happens at or above it?
- Which four behavior signals were strongest in the model review?
- When is the pilot planned to start and when is the formal review?
- What are the raw staging extract retention and access-log retention periods?
- What should happen if the feature snapshot is older than 30 hours?
- Which fields are explicitly excluded from the model input?
- What information is not specified about the personal identity of any real customer?

End of report. This is a fictional educational sample for testing document extraction and RAG question answering. All data, organizations, teams, dates, performance figures, and examples are synthetic.